

From council tax to land value tax: who gains from a revenue-neutral reform?*

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Abstract

Council tax, the UK’s principal recurrent property tax, rests on 1991 valuations and is regressive with respect to property value. We use PolicyEngine UK, an open-source static microsimulation model built on the Enhanced Family Resources Survey, to simulate replacing council tax’s £57.6 billion of net revenue with a flat annual tax on the unimproved value of land, imputing land to every household from Wealth and Assets Survey property wealth, region-specific land shares derived from MHCLG land value estimates, and an allocation of corporate land. Against a land base of £7.46 trillion the budget-neutral rate is 0.77 per cent. We find that 68 per cent of households gain and that absolute poverty after housing costs falls by 0.70 percentage points, while poverty before housing costs and the Gini coefficient of equivalised net income are both essentially unchanged. The central result is a dissociation between the two ways of ranking households. Across income deciles the reform is close to distributionally flat, with average effects in deciles 2–8 of a few hundred pounds either way and clear losses confined to the top two deciles. Across wealth deciles the gradient is monotonic and steep: the bottom seven gain, and the top decile — which holds 47 per cent of all land — loses £5,752 a year, with 98 per cent of its households worse off. Because land is highly concentrated by wealth but only weakly correlated with current income, income-based distributional statistics are close to uninformative about this reform. The pattern survives changes to the replacement target, the treatment of corporate and foreign-owned land, the geographic scope and plausible perturbations of the land shares; full capitalisation would imply a one-off fall in land values of 15–26 per cent, borne overwhelmingly by the top wealth decile.

Keywords: land value tax, council tax, microsimulation, tax incidence, poverty, inequality, United Kingdom

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1 Introduction

Council tax is the United Kingdom’s principal recurrent tax on residential property, with gross receipts of approximately £62 billion forecast for 2026–27 (OBR, 2025). Its design has attracted sustained criticism in the public-finance literature (Mirrlees et al., 2011; Adam et al., 2020). Dwellings in England and Scotland remain assessed on their estimated value as at 1 April 1991; the banding structure is capped, with the top band in England (Band H) covering all properties worth £320,000 or more *in 1991 prices*, so that properties of widely differing current values share the top band and face nearly identical bills; and effective tax rates decline as a share of property value as that value rises. Council tax is therefore regressive with respect to the value of the asset it nominally taxes, bears a weak relationship to households’ current ability to pay, and discourages investment in property improvements.

This paper examines the household-level consequences of abolishing council tax and replacing its net revenue, pound for pound, with a flat annual tax on the unimproved value of land. The land value tax (LVT) has a long intellectual pedigree, from George (1879) through the optimal-taxation literature (Ramsey, 1927) to the review by Mirrlees et al. (2011), resting on the observation that the supply of land is fixed: a tax on pure land rents raises revenue without discouraging productive activity. The UK policy debate, however, has to date lacked a household-level quantification of the gains and losses such a reform would generate.

We provide such a quantification. Using PolicyEngine UK, an open-source static microsimulation model of the UK tax and benefit system built on an enhanced version of the Family Resources Survey (PolicyEngine, 2026), we proceed in three steps. First, we impute a land value to every household by combining survey property wealth with region-specific land shares and an allocation of corporate-sector land, calibrating national aggregates to the ONS National Balance Sheet (ONS, 2025). Second, we reconstruct council tax liabilities bottom-up by region, band and discount status, calibrated to Valuation Office Agency dwelling counts. Third, we simulate replacing the model’s net council tax revenue of £57.6 billion (gross liabilities of £61.9 billion less £4.3 billion of council tax reduction) with a flat-rate LVT for fiscal year 2026–27; Section 3 discusses how this model-based figure compares with the OBR receipts forecast. Against an estimated UK land base of £7.46 trillion, we calculate that revenue neutrality requires a rate of 0.77 per cent of land value per year.

We report three principal findings. First, most households gain: 68.2 per cent are better off, and the poorest income decile gains £647 per year, 3.0 per cent of its net income, because its average council tax bill of £1,383 is nearly twice the £736 land tax it would owe. Second, the effect across the income distribution is otherwise close to flat and imprecisely estimated: average changes in deciles 2–8 range from –£163 to +£578 with no clear gradient, and only deciles 9 and 10 record substantial losses (£1,016 and £963). Absolute poverty after housing costs falls by 0.70 percentage points, but poverty before housing costs is unchanged (+0.04 points), as is the Gini coefficient of equivalised net income (+0.0003). Third, and in sharp contrast, the effect across the *wealth* distribution is monotonic and large: wealth deciles 1–7 gain between £523 and £1,480, decile 9 loses £1,069, and the top wealth decile — which holds 47.4 per cent of all land — loses £5,752 per year, 8.8 per cent of its net income, with 98.4 per cent of its households worse off.

The three findings are reconciled by the joint distribution of land and income. Land is far more unequally held than income (we estimate a wealth Gini of 0.70 against 0.36 for income) but is only weakly correlated with current income, so a tax on land redistributes sharply across the wealth distribution while barely registering in income-based inequality statistics. This is

a substantive point about measurement as much as about land: a reform of this kind can be evaluated as approximately distribution-neutral on the statistics that conventionally govern UK tax debate, and as strongly redistributive on the dimension it actually operates on.

In addition to the central budget-neutral scenario, we trace out a rate grid from 0.5 to 5 per cent, documenting how revenue (from £37.3 billion to £373.2 billion), poverty and inequality vary with the rate; we decompose the tax base into household and corporate land; we verify that the swap generates no interactions with means-tested benefits, so that the household-level effect is exactly the abolished council tax bill less the land tax bill; and we test the results against alternative replacement targets, bases and geographic scope, against a proportional property tax comparator, and against full capitalisation of the levy into land prices.

The remainder of the paper proceeds as follows. Section 2 situates the analysis in the literature on land taxation and UK property-tax reform. Section 3 describes the data, the land-value imputation, the council tax baseline and the reform implementation. Section 4 presents the distributional results. Section 5 summarises the findings, discusses limitations and policy implications, and concludes. An appendix reports the full rate grid, a summary of landless households, a worked single-household example and a data and parameter summary.

2 Related literature

2.1 The economics of land value taxation

The case for taxing land rests on the classical observation that land is supplied inelastically. [George \(1879\)](#), in *Progress and Poverty*, argued that the rental value of unimproved land is a socially created surplus — driven by population, public infrastructure and agglomeration rather than by any effort of the owner — and proposed a “single tax” on land values to replace taxes on labour and capital. The modern optimal-taxation literature formalises the intuition: because the base cannot shrink in response to the tax, a levy on pure land rents generates no substitution effect and hence no deadweight loss, making it the canonical example of efficient taxation in the [Ramsey \(1927\)](#) inverse-elasticity framework. [Feldstein \(1977\)](#) qualified the classical view by noting general-equilibrium portfolio effects, but the presumption that land taxes are unusually efficient survives.

Two further strands reinforce the theoretical case. [Arnott and Stiglitz \(1979\)](#) establish the Henry George Theorem: under conditions of optimal city size, aggregate land rents exactly finance the efficient level of local public expenditure, giving land taxation a normative role beyond efficiency — it is the natural fiscal counterpart of the public goods that generate land value. In cross-country empirical work, [Johansson et al. \(2008\)](#) rank recurrent taxes on immovable property as the least harmful to long-run growth among the major tax instruments, ahead of consumption, personal income and corporate income taxes. More recently, [Bonnet et al. \(2021\)](#) document that the post-1980 rise in wealth-to-income ratios is to a large extent a rise in the price of residential land, and argue that land is accordingly both an increasingly important and an administratively feasible tax base for contemporary economies.

The empirical evidence, while limited by the scarcity of implemented land taxes, is broadly consistent with the theory. [Oates and Schwab \(1997\)](#) examine Pittsburgh’s two-rate property tax — which taxed land at more than five times the rate applied to structures — and find that the city sustained substantially higher building activity than comparable cities in the same period; they attribute this partly to the land tax’s neutrality with respect to development, while noting the difficulty of isolating the tax’s contribution from concurrent local conditions.

Tideman (1995) argues on theoretical grounds that land taxation is better than neutral, as it discourages speculative withholding of developable land. The contributions collected in Dye and England (2009) temper this optimism by documenting the practical assessment challenges — above all, separating land from improvement value at scale — that have historically limited adoption.

2.2 UK property taxation and council tax

The deficiencies of council tax are well documented. Mirrlees et al. (2011) concluded that UK housing taxation lacks a coherent rationale: council tax is regressive in property value and based on obsolete 1991 valuations, while stamp duty land tax taxes transactions and thereby impedes mobility. The review proposed replacing both with a Housing Services Tax proportional to the consumption value of housing, complemented by a land value tax on business land — an explicit endorsement of land taxation where valuation is feasible. Subsequent work has quantified the regressivity that motivates reform: Adam et al. (2020) show that effective council tax rates as a share of property value fall sharply across the price distribution, and that the absence of revaluation in England and Scotland leaves relative price changes since 1991 — most notably London’s appreciation — untaxed. Muellbauer (2005) argues that reforming council tax into a proportional property tax with regular automatic revaluation would also stabilise the housing market, and the OECD’s most recent survey of the UK likewise recommends updating property valuations and reforming council tax (OECD, 2024).

A body of applied UK work has modelled alternatives. Corlett and Gardiner (2018) examine options including a proportional property tax and a partial LVT at the aggregate and regional level; Fairer Share (2021) propose a proportional property tax of 0.48 per cent of current property value and estimate that roughly three-quarters of households would pay less — a useful comparator for our finding that 68 per cent of households gain at a 0.77 per cent tax on the narrower land base. Murphy and Snelling (2019) model council tax reform for London. In Scotland, the cross-party Commission on Local Tax Reform (Scottish Government, 2015) concluded that the present system must end and assessed property, land and income bases, and Hughes et al. (2018) examine LVT options for the Scottish Land Commission, highlighting valuation and transition as the binding constraints. Leunig (2024) propose replacing both council tax and stamp duty with proportional levies on current values. Closest to our exercise, ap Gwilym et al. (2020) assess the feasibility of a local land value tax for Wales for the Welsh Government, simulating revenue-neutral rates at local-authority level using small-area land value estimates.

2.3 Contribution

Relative to this literature, our contribution is a household-level microsimulation of a full council-tax-to-LVT swap for the UK, with land values imputed to every household in a nationally representative micro-dataset, benchmarked against national balance-sheet aggregates, and evaluated within a complete model of the tax and benefit system. Prior UK LVT analyses have generally worked with regional, band-level or local-authority aggregates — the Welsh assessment of ap Gwilym et al. (2020) being the most detailed to date — rather than with household microdata. The microdata buy us three things that aggregate analyses cannot deliver: the joint distribution of land and income, which drives the winners-and-losers results and the divergence between income-ranked and wealth-ranked incidence; poverty rates measured against the statutory equivalised, housing-cost-adjusted thresholds; and a decomposition of who bears the

corporate component of the base. We note in Section 4 that the swap turns out to generate no interactions with means-tested benefit entitlements, so the household-level net income change is exactly the difference between the abolished council tax bill and the new land tax bill — a finding, not an assumption, and one that materially simplifies interpretation. To our knowledge this is the first open-source, household-level distributional analysis of a revenue-neutral LVT for the whole of the UK, and the code and data pipeline are fully reproducible (PolicyEngine, 2026).

3 Data and methodology

3.1 Microdata

Our simulations use PolicyEngine UK (PolicyEngine, 2026), an open-source static microsimulation model of the UK tax and benefit system, run on the Enhanced Family Resources Survey (EFRS) for 2023–24, uprated to fiscal year 2026–27. The EFRS augments the standard Family Resources Survey in two ways. First, because household surveys under-represent the top of the income distribution, synthetic high-income records are generated by training a quantile random forest on HMRC’s Survey of Personal Incomes (SPI) and imputing the upper tail of incomes. Second, household weights are calibrated by gradient descent so that the weighted dataset matches a large set of national and regional administrative totals from the ONS, HMRC and DWP — including population by age and region, income components, and benefit caseloads. Wealth variables, including property wealth and corporate (financial) wealth, are imputed from the ONS Wealth and Assets Survey (WAS).

3.2 Council tax baseline

The FRS records council tax imperfectly, so we reconstruct liabilities. Each household is assigned a council tax band (A–H in England and Scotland, A–I in Wales) and a bill equal to the average band rate in its region, with the 25 per cent single-person discount imputed to single-adult households. Band assignment probabilities are calibrated so that the distribution of dwellings by region and band matches Valuation Office Agency (and Scottish Assessors) counts. This bottom-up reconstruction yields, for 2026–27, gross council tax of £61.9 billion; netting off £4.3 billion of means-tested council tax reduction gives net revenue of

$$R_{CT} = 61.9 - 4.3 = \text{£}57.6 \text{ billion}, \quad (1)$$

which is the amount the land value tax must replace.

We emphasise that this is a modelled liability total, calibrated to dwelling counts by region and band, and not a receipts figure. It exceeds the Office for Budget Responsibility’s accruals-based council tax receipts forecast of approximately £53.7 billion (OBR, 2025) by some 7 per cent of net revenue. Part of the gap is conceptual — receipts are net of collection losses, discounts and exemptions that a liability model applied to average band rates does not fully reproduce — and part reflects genuine calibration error in the band-rate averages. Because the headline rate is the ratio of this figure to the land base, we report throughout a second replacement target calibrated to OBR receipts, which implies a revenue-neutral rate of 0.72 rather than 0.77 per cent; Section 4.7 reports the distributional results under both. The 0.77 per cent central case has the advantage of internal consistency: the simulated LVT replaces exactly the council tax the model abolishes, household by household, so that aggregate net income is unchanged by

construction. Northern Ireland operates a separate domestic rates system rather than council tax; our central specification leaves it unchanged, and Section 4.7 reports results excluding Northern Ireland.

3.3 Imputing land values

The tax base is the unimproved value of land held directly (residential land under owner-occupied and other household-owned property, plus any land held as an asset in its own right) and indirectly (land on corporate balance sheets, attributed to households as ultimate owners). For household i in region r , land value is

$$L_i = \underbrace{\lambda_r P_i + A_i}_{\text{household land } H_i} + \underbrace{\frac{K_i}{\sum_j K_j w_j} L^{\text{corp}}}_{\text{allocated corporate land } C_i}, \quad (2)$$

where P_i is WAS-imputed property wealth, λ_r is the region-specific land share of property value, A_i is directly owned land reported separately in the WAS asset schedule (small in aggregate), K_i is the household's corporate wealth, w_j are survey weights, and L^{corp} is aggregate corporate-sector land. Corporate land is allocated proportionally to each household's corporate wealth, treating shareholdings (direct and via pensions) as the ultimate claim on corporate assets. Note that the allocation is normalised to the weighted population total, so $\sum_i C_i w_i = L^{\text{corp}}$ holds exactly by construction; the corporate component of the base is therefore pinned to its aggregate, whereas the household component is not.

Regional land shares. The shares λ_r are derived by the residual method from the MHCLG *Land Value Estimates for Policy Appraisal* (MHCLG, 2023) and the UK House Price Index. For each English local authority we take the MHCLG central-scenario residential land value per hectare, divide by dwellings per hectare to obtain land value per dwelling, and divide that by the average house price, giving

$$\lambda_r = \frac{\text{MHCLG land value per hectare}_r / \text{dwellings per hectare}_r}{\text{average house price}_r}, \quad (3)$$

aggregated from local authority to region. Because MHCLG covers England only, Scotland and Wales are interpolated from the English regions closest in house price (Scotland from the North East and North West, Wales from the North West), and Northern Ireland is assigned the population-weighted mean. Table 1 reports the resulting values. They range from 85 per cent in London to 42 per cent in the North East, reflecting the fact that in high-price regions almost all of the price differential is location value rather than structure cost, while in low-price regions the price approaches the cost of the structure. The interpolation for the devolved nations is a material assumption — roughly 13 per cent of GB dwellings are assigned a land share estimated from English data — and we test sensitivity to it in Section 4.7.

Aggregates. The corporate aggregate L^{corp} is taken from the sectoral tables underlying the ONS National Balance Sheet (ONS, 2025), which give UK land of £7.10 trillion at end-2024, split £5.04 trillion household and £2.06 trillion corporate. (These derived series differ from the headline bulletin presentation, which reports household non-produced assets of approximately £4.6 trillion in 2024 on a different sectoral grouping; the household/corporate split is applied

Table 1: Regional land share of property value, λ_r

Region	λ_r	Basis
London	0.85	MHCLG, England
South East	0.65	MHCLG, England
East of England	0.60	MHCLG, England
South West	0.58	MHCLG, England
West Midlands	0.52	MHCLG, England
East Midlands	0.48	MHCLG, England
North West	0.47	MHCLG, England
Wales	0.47	Interpolated (North West)
Yorkshire and the Humber	0.46	MHCLG, England
Scotland	0.44	Interpolated (North East / North West)
North East	0.42	MHCLG, England
Northern Ireland	0.67	Population-weighted mean

Source: Equation (3), MHCLG *Land Value Estimates for Policy Appraisal 2023* central scenario and UK House Price Index; implemented in PolicyEngine UK as

`household.wealth.land.intensity.property.wealth.by.region`.

using the last directly observed sectoral shares, from 2020.) The corporate series is held at its 2024 level for 2025 and 2026, so $L^{\text{corp}} = \mathcal{L}2.06$ trillion enters the simulation unchanged.

The household aggregate is *not* imposed. It emerges from equation (2) applied to EFRS property wealth, which is itself uprated from the 2023–24 survey to 2026–27 by the model’s house-price and financial-asset uprating indices. The weighted household total is

$$H = \sum_i H_i w_i = \mathcal{L}5.40 \text{ trillion}, \quad (4)$$

which is 107.1 per cent of the corresponding un-uprated 2024 ONS household benchmark of $\mathcal{L}5.04$ trillion. Roughly speaking, the 7 per cent excess is the two years of nominal property-price growth embodied in the uprated microdata, with the residual reflecting imputation error in WAS property wealth and the fact that the regional household-land targets are not binding constraints in the weight calibration. The total base is therefore

$$L = H + L^{\text{corp}} = 5.40 + 2.06 = \mathcal{L}7.46 \text{ trillion}, \quad (5)$$

or 105.1 per cent of the un-uprated 2024 ONS total. We report this as a model output benchmarked against ONS rather than as a calibrated target, and Section 4.7 shows how the results move if the household base is instead scaled to the ONS 2024 level. Household land so measured is 68.4 per cent of total property wealth and total land is 27.7 per cent of total household wealth ($\mathcal{L}26.98$ trillion); mean land value per household, including allocated corporate land, is $\mathcal{L}232,672$ and the median is $\mathcal{L}93,796$.

3.4 The reform and the revenue-neutral rate

The reform (i) abolishes council tax in Great Britain and (ii) levies a flat annual tax at rate r on each household’s land value L_i , implemented in PolicyEngine UK via the parameters

`gov.contrib.ubi_center.land_value_tax.rate` and `gov.contrib.abolish_council_tax`. The budget-neutral rate equates LVT revenue with net council tax revenue:

$$r^* = \frac{R_{CT}}{L} = \frac{57.6}{7,463} = 0.77\%. \quad (6)$$

The net income change for household i is $\Delta y_i = CT_i - r^* L_i$, where CT_i is the abolished council tax liability (net of council tax reduction). Three maintained assumptions govern incidence, all standard in static microsimulation. First, incidence is assigned entirely to current landowners: we assume no behavioural response and no capitalisation of the tax into land prices. Second, rents are held fixed, so renters — who own no residential land — retain their abolished council tax liability in full and pay LVT only on any corporate wealth they hold. Third, the land base is held fixed across the rate grid, consistent with the inelastic supply of land. All results are for simulation year 2026, with poverty measured as absolute poverty against a baseline-fixed threshold, before housing costs (BHC) and after housing costs (AHC), and inequality by the Gini coefficient of equivalised household net income. In addition to the central rate $r^* = 0.77\%$ we simulate a grid of rates $\{0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$ per cent to trace out revenue and distributional effects, and decompose revenue by base scope (household-only versus corporate-only land).

4 Results

4.1 Where land value sits

The distributional consequences of an LVT are determined by the distribution of its base, which we describe first. Table 2 reports average land value, property wealth and land shares by decile of equivalised household net income. Two features stand out. First, average land value rises with income, but weakly and unevenly: it is £95,302 in the poorest decile and £442,935 in the richest, a ratio of 4.6 to 1 against an income ratio of 6.5 to 1, and the profile is far from monotonic — decile 2 (£217,644) holds nearly twice as much land on average as decile 3 (£115,702). Some of this reflects genuine heterogeneity, principally asset-rich retired households with low current income; some reflects imputation noise in a base constructed from survey property wealth. Either way, the income gradient of the tax base is shallow, which is what drives the results in Section 4.6. Second, the top two income deciles together hold 35.5 per cent of land, whereas, as Section 4.4 shows, the top two *wealth* deciles hold 66.6 per cent.

Table 2 separates the two components of the base, which are not comparable quantities: household land is a share of the household’s own property wealth, while allocated corporate land is an attributed claim on corporate balance sheets that households holding no property may still bear. Reporting them together against property wealth — as a single “land value” column would — can make land appear to exceed the property wealth it is derived from.

Regional variation is at least as pronounced as variation by income. Average household land value is £369,802 in London and £286,311 in the South East, compared with £153,631 in the North East and £159,535 in Scotland (Figure 2) — a ratio of 2.4 to 1 that reflects both higher property prices and higher land shares of those prices in the South. Across household types, pensioner couples hold the most land on average (£431,377), followed by working-age couples without children (£303,440); lone parents, who are predominantly renters, hold very little (£7,797) (Figure 3). Ranked by total wealth rather than income, concentration is far greater: the bottom three wealth deciles hold essentially no land, while the top wealth decile

Table 2: Land value and property wealth by decile of equivalised household net income, 2026–27

Income decile	Avg. income (£/yr)	Avg. household land (£)	Avg. corporate land (£)	Avg. total land (£)	Avg. property wealth (£)	Share of land (%)
1	21,701	75,848	19,454	95,302	106,712	4.7
2	31,068	171,344	46,300	217,644	198,309	8.8
3	37,182	89,340	26,362	115,702	144,651	5.0
4	38,400	152,099	50,164	202,263	204,378	9.3
5	43,127	144,852	52,305	197,157	213,009	9.0
6	53,985	160,086	50,458	210,544	231,152	8.5
7	55,357	138,281	48,129	186,410	221,029	8.1
8	69,125	191,548	88,673	280,220	308,960	10.9
9	77,096	294,523	127,118	421,641	451,521	17.5
10	141,327	294,942	147,992	442,935	429,497	18.0

Source: Authors’ calculations using PolicyEngine UK.

averages £1,088,892 — 47.4 per cent of all land (Table 5 below). Overall, 17.2 per cent of households hold no land.

4.2 Revenue and the rate schedule

Because the base is fixed in the static framework, LVT revenue is linear in the rate: each 0.1 percentage point raises approximately £7.5 billion. Table 3 reports revenue across the rate grid. A 0.5 per cent rate raises £37.3 billion — £20.3 billion less than council tax — while a 1 per cent rate raises £74.6 billion and a 5 per cent rate raises £373.2 billion. The budget-neutral rate is 0.77 per cent (equation 6). Decomposing the base at a 1 per cent rate, household land contributes £54.0 billion and corporate land £20.6 billion of the £74.6 billion total. The corporate component — 28 per cent of revenue — falls on households in proportion to their shareholdings, and its inclusion permits a lower rate for any given revenue target than a purely residential base would require.

Table 3: LVT revenue by rate, 2026–27

Rate	LVT revenue (£bn)	Council tax (£bn)	Net revenue (£bn)	Avg. LVT per household (£)
0.5%	37.3	57.6	−20.3	1,163
1.0%	74.6	57.6	17.0	2,327
1.5%	111.9	57.6	54.3	3,490
2.0%	149.3	57.6	91.6	4,653
3.0%	223.9	57.6	166.3	6,980
5.0%	373.2	57.6	315.5	11,634

Note: Net revenue is LVT revenue less replaced council tax revenue. *Source:* Authors’ calculations using PolicyEngine UK.

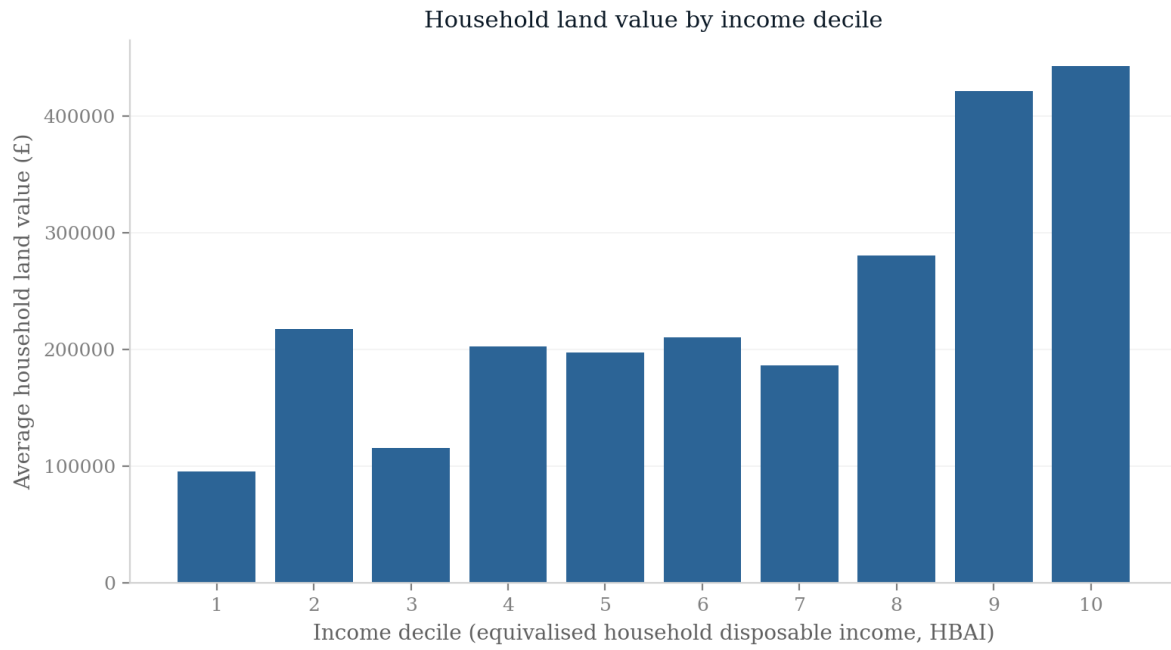


Figure 1: Average land value by income decile.

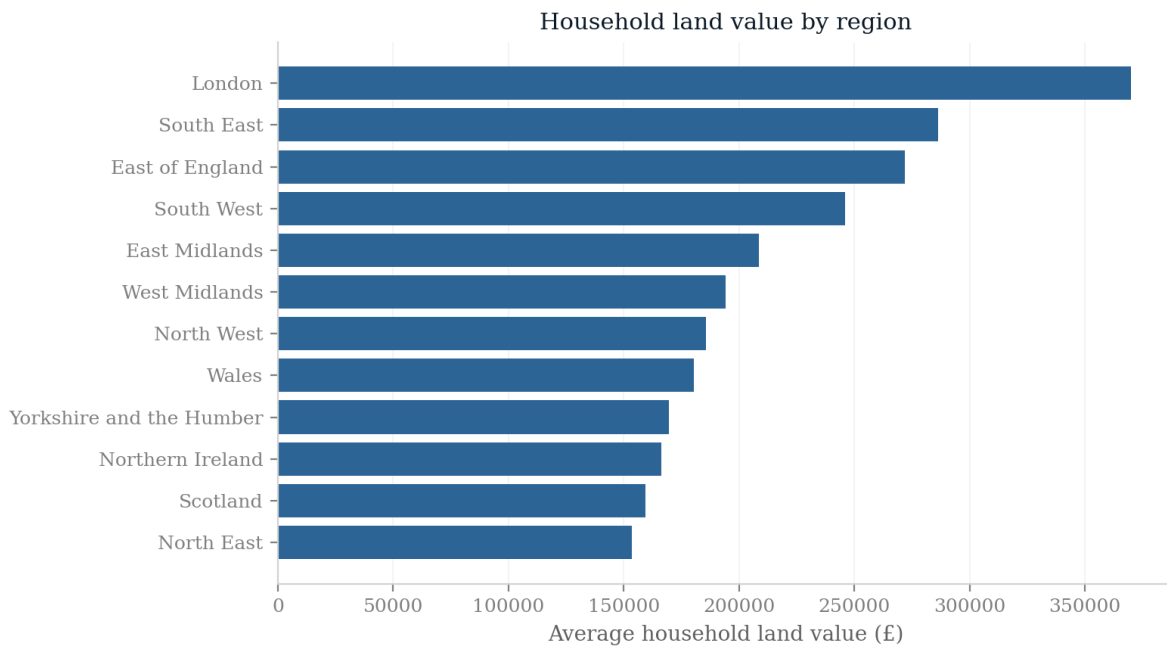


Figure 2: Average household land value by region.

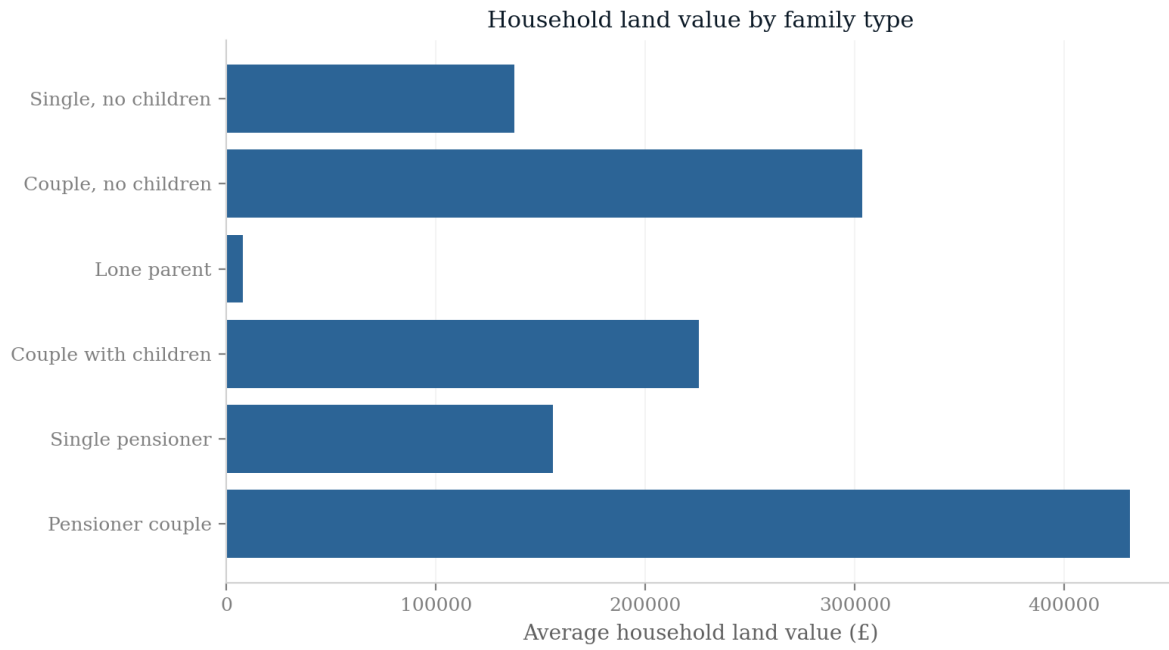


Figure 3: Average household land value by family type.

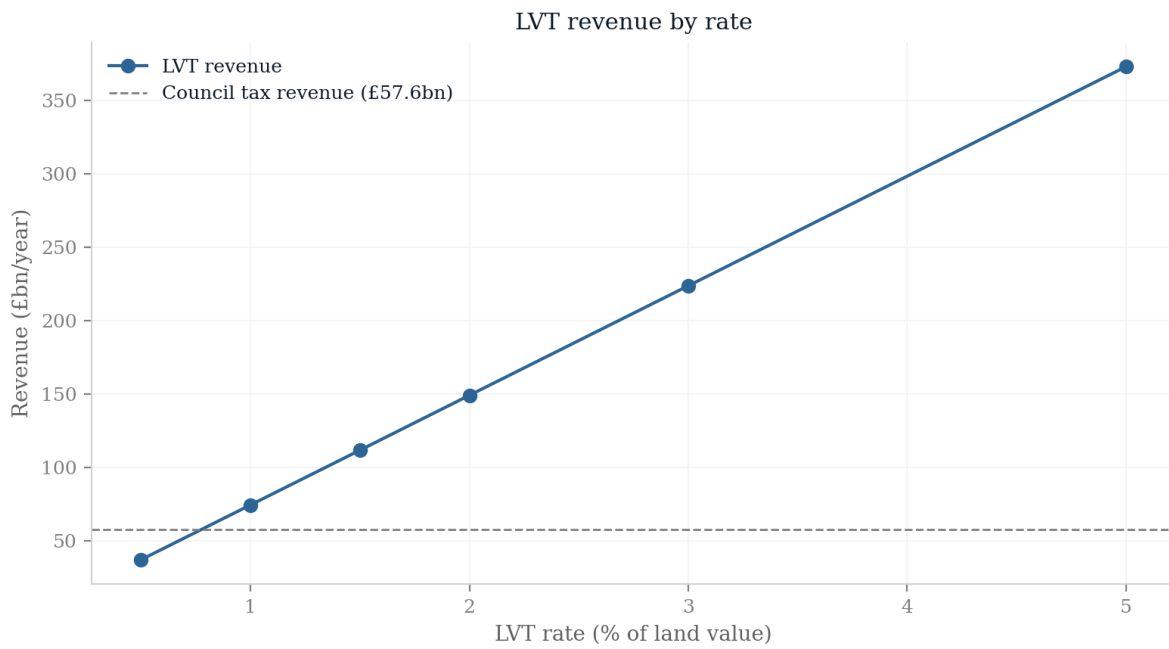


Figure 4: LVT revenue by rate against the council tax replacement target.

4.3 Distributional impact by income decile

Table 4 and Figure 5 report the average change in household net income at the budget-neutral 0.77 per cent rate, by income decile. Because the swap generates no benefit interactions — a result we verify in Section 4.7 — the change is exactly the abolished council tax bill less the land tax bill, $\Delta y_i = CT_i - r^* L_i$.

The pattern across income deciles is best described as flat with a step down at the top. The poorest decile gains most in both absolute and relative terms, £647 per year or 3.0 per cent of net income, because its average council tax bill of £1,383 — inflated relative to its means by the compressed banding structure — is nearly twice the £736 land tax charged on its modest holdings. Deciles 3 and 7 gain £578 and £455; deciles 5, 6 and 8 are close to unchanged; and deciles 2 and 4 record small average *losses* of £163 and £60, reflecting the uneven land profile documented in Table 2 rather than anything systematic about those income levels. Only in deciles 9 and 10 is the loss substantial and clearly attributable to income: £1,016 and £963 per year, as land holdings of £420,000–£443,000 attract liabilities of £3,250–£3,420 against council tax savings of £2,240–£2,460. In percentage terms the loss is larger for decile 9 (−1.3 per cent) than for decile 10 (−0.7 per cent), because top-decile incomes are considerably higher while average land holdings are only slightly larger.

We stress that the middle of this profile is not precisely estimated. The differences between adjacent deciles in the range 2–8 are of the order of a few hundred pounds a year on average net incomes of £31,000–£69,000, and they rest on imputed land values whose decile means are visibly noisy. The result that should be read from Table 4 is that the reform is roughly distributionally neutral across the bottom eight deciles taken together and clearly costly to the top two — not that any particular middle decile gains or loses.

Table 4: Impact of the 0.77% budget-neutral LVT by income decile

Income decile	Avg. LVT (£)	Avg. CT saved (£)	Avg. net change (£)	% of net income	Winners (%)	Losers (%)
1	736	1,383	+647	+3.0	79.0	18.3
2	1,681	1,518	−163	−0.5	72.0	26.9
3	894	1,472	+578	+1.6	76.5	22.4
4	1,562	1,502	−60	−0.2	72.6	26.6
5	1,523	1,668	+145	+0.3	69.1	29.7
6	1,626	1,805	+179	+0.3	70.9	28.6
7	1,440	1,895	+455	+0.8	76.0	23.7
8	2,164	2,199	+35	+0.1	67.9	30.9
9	3,256	2,240	−1,016	−1.3	48.1	51.4
10	3,421	2,458	−963	−0.7	48.6	50.8

Note: “CT saved” denotes the abolished council tax liability net of council tax reduction. Winners and losers do not sum to 100 per cent because a small share of households are unaffected. *Source:* Authors’ calculations using PolicyEngine UK.

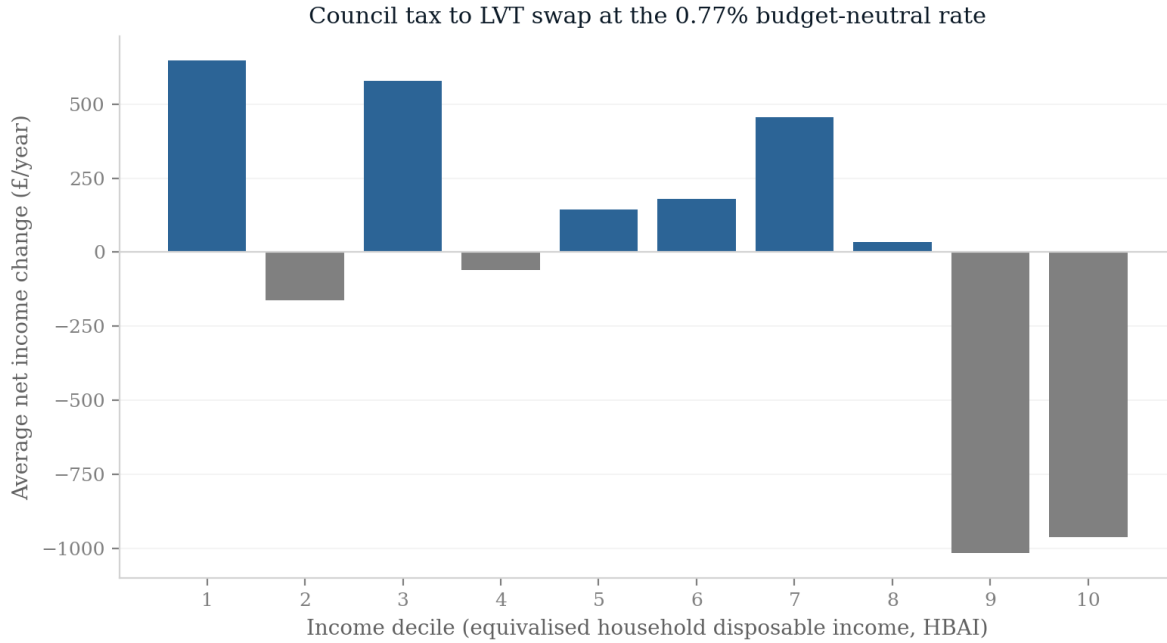


Figure 5: Average net income change by income decile at the 0.77% budget-neutral rate.

4.4 Distributional impact by wealth decile

Table 5 and Figure 6 report the corresponding results when households are ranked by total wealth rather than income.¹ The gradient here is monotonic and steep, in sharp contrast to the income ranking. The bottom two wealth deciles own essentially no land and retain their council tax savings in full, gaining £898 and £902 per year; deciles 3–7 gain between £523 and £1,480; decile 8 loses £268; decile 9 loses £1,069; and the top wealth decile — averaging £1.09 million of land — loses £5,752 per year, or 8.8 per cent of its net income, with an average LVT liability of £8,410 against council tax savings of £2,658. In the top wealth decile 98.4 per cent of households lose; in each of the bottom seven, more than 82 per cent gain. The incidence of the reform tracks land wealth almost mechanically, which is exactly what the instrument is designed to do.²

4.5 Winners and losers within deciles

Decile averages conceal substantial within-decile variation, because land holdings vary widely within any given income band: an outright owner-occupier in London and a private renter in the North East may occupy the same income decile. At the budget-neutral rate, 68.2 per cent of all

¹We construct equal-weight wealth deciles directly. PolicyEngine UK’s own `household_wealth_decile` variable is degenerate at the bottom of the distribution — the mass of households with zero or negative net wealth is tied, leaving decile one empty and placing 22.6 per cent of households in decile two — which would make the decile averages in this table incomparable.

²Revenue neutrality holds in aggregate — LVT revenue equals abolished council tax revenue at £57.6 billion — so the population-weighted average net-income change is zero. It is identical whether households are grouped by income or by wealth decile, since the two tables show the same households re-sorted; the per-decile averages need not sum to zero because decile group sizes are equal in population but not in the underlying quantities.

Table 5: Impact of the 0.77% budget-neutral LVT by wealth decile

Wealth decile	Avg. land value (£)	Avg. LVT (£)	Avg. CT saved (£)	Avg. net change (£)	Winners (%)	Losers (%)
1	1	0	898	+898	82.1	13.7
2	29	0	902	+902	83.2	11.4
3	1,265	10	1,177	+1,168	91.2	8.1
4	25,263	195	1,675	+1,480	94.4	5.6
5	76,699	592	1,833	+1,241	91.8	8.2
6	138,998	1,074	2,027	+953	89.5	10.5
7	212,347	1,640	2,163	+523	82.9	17.1
8	321,469	2,483	2,215	−268	48.0	52.0
9	451,386	3,486	2,417	−1,069	20.0	80.0
10	1,088,892	8,410	2,658	−5,752	1.6	98.4

Note: Deciles of total household wealth. *Source:* Authors' calculations using PolicyEngine UK.

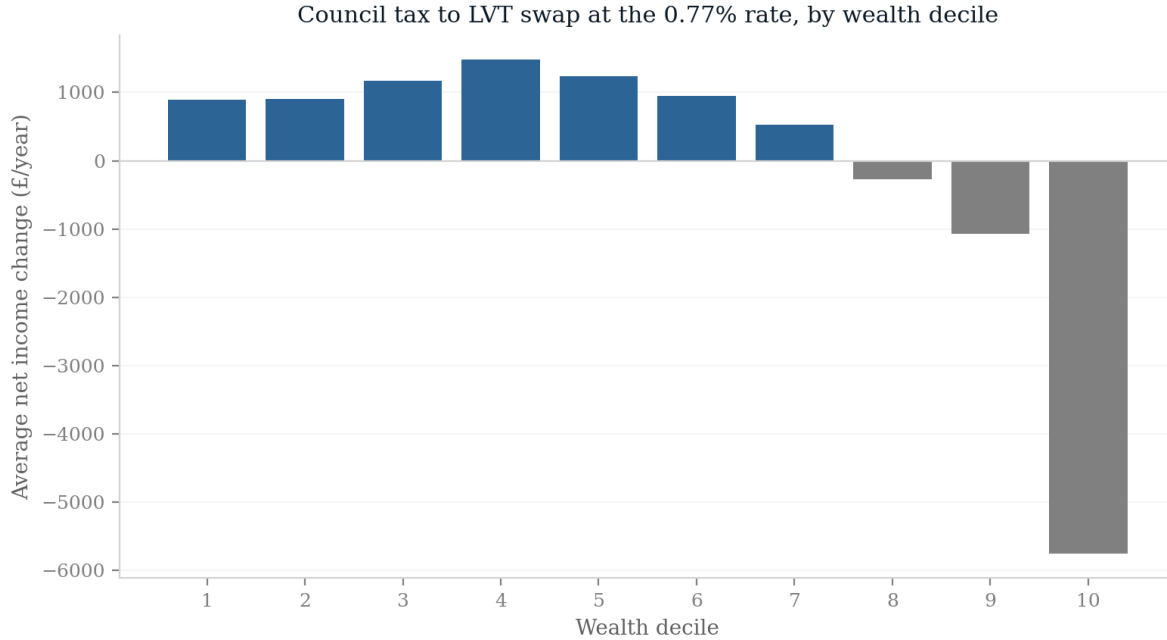


Figure 6: Average net income change by wealth decile at the 0.77% budget-neutral rate.

households gain. The share of gaining households declines only gradually with income through deciles 1–8 (from 79.0 to 67.9 per cent) before falling to just under one half in deciles 9–10 (Figure 7). Even in the top income decile, 48.6 per cent of households gain — predominantly high earners who rent or own property of modest value — while 18.3 per cent of bottom-decile households lose, typically low-income owners of valuable land. Sorted by wealth, the gradient is considerably steeper: the share of gaining households exceeds 82 per cent in each of the bottom seven wealth deciles and falls to 1.6 per cent in the top decile. The 17.2 per cent of households holding no land gain £832 per year on average, and 81.5 per cent of them gain; the remainder are largely households whose council tax was already fully offset by council tax reduction.

That the winner share is high while the average income-decile effects are small is not a contradiction. The reform moves a fixed £57.6 billion from a base that is broad and shallow (council tax, capped at the top) to one that is narrow and deep (land, highly concentrated). Most households pay a little less; a small minority pay a great deal more. This is visible in the wealth ranking and almost invisible in the income ranking.

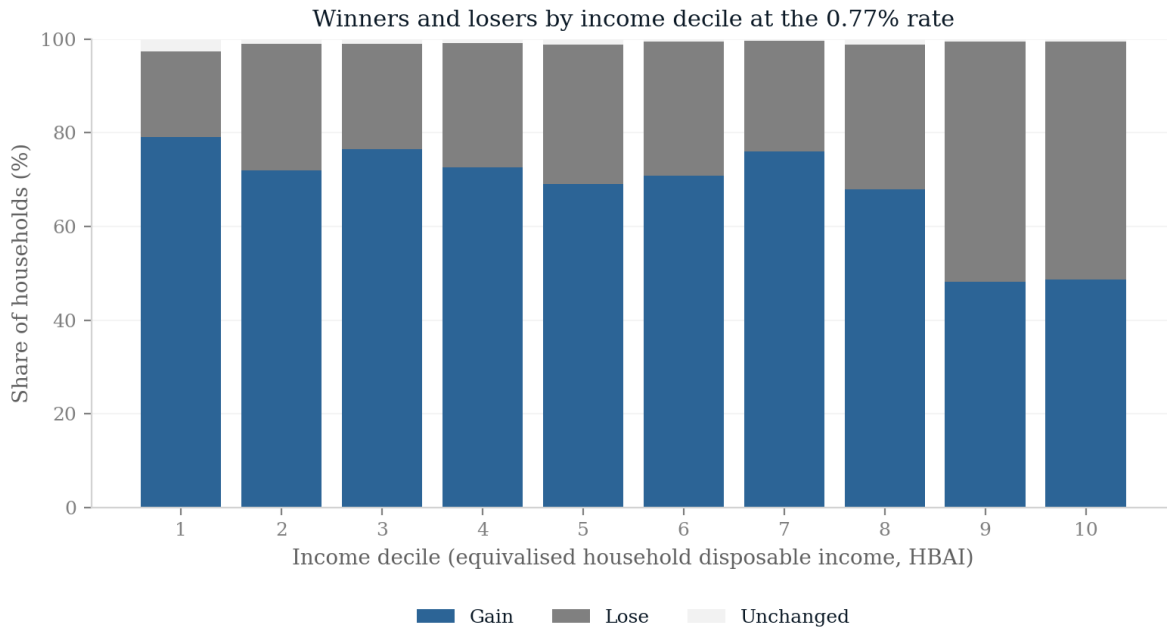


Figure 7: Share of winners and losers by income decile at the 0.77% budget-neutral rate.

4.6 Poverty and inequality

Table 6 reports poverty and inequality across the rate grid, measured against baseline-fixed absolute poverty thresholds. At the budget-neutral 0.77 per cent rate the two poverty measures move in opposite directions. Absolute poverty after housing costs falls from 14.75 to 14.04 per cent, a reduction of 0.70 percentage points, while poverty before housing costs is essentially unchanged, edging up from 11.53 to 11.58 per cent (+0.04 points). The Gini coefficient of equivalised household net income is likewise unchanged, rising by 0.0003 from a baseline of 0.3641.

The BHC and AHC divergence is informative rather than a puzzle. The AHC measure nets housing costs out of income, and the households lifted above the AHC line are disproportionately low-income renters, who gain their entire council tax bill and owe almost no land tax. The BHC measure does not, so it gives more weight to low-income owner-occupiers, who lose. The near-zero BHC effect is the sum of these two flows, and it is small enough that we would not claim a poverty reduction on the BHC measure at all: the honest statement is that the revenue-neutral swap reduces after-housing-cost poverty modestly and leaves before-housing-cost poverty unchanged.

The grid shows how sensitive this is to the rate, and therefore to the replacement target. At 0.5 per cent — which is not revenue-neutral, reducing net revenue by £20.3 billion — poverty falls on both measures (BHC -0.60 , AHC -0.91 points) and the Gini declines slightly (-0.0028). At 1 per cent, which raises £17.0 billion of net revenue, BHC poverty rises 0.83 points while AHC poverty is still 0.30 points below baseline. At higher rates poverty and measured income inequality rise substantially — at 5 per cent, BHC poverty rises by 12.0 points and the Gini by 0.10 — because the static analysis levies cash charges on asset-rich, income-poor households with no offsetting use of the additional revenue. These upper-grid rows are the gross incidence of the tax alone and not complete reform packages: any practical proposal at those rates would recycle the net revenue, and the rows should not be read as forecasts of what such a package would do. The wealth Gini (0.7038) is mechanically unchanged, since the simulation does not model asset-price responses; Section 4.7 reports what capitalisation would imply for it.

Table 6: Poverty and inequality by LVT rate (council tax abolished throughout)

Rate	Poverty BHC (%)	Δ BHC (pp)	Poverty AHC (%)	Δ AHC (pp)	Δ Gini
Baseline	11.53	—	14.75	—	—
0.5%	10.94	-0.60	13.84	-0.91	-0.0028
0.77%	11.58	$+0.04$	14.04	-0.70	$+0.0003$
1.0%	12.36	$+0.83$	14.45	-0.30	$+0.0033$
1.5%	13.82	$+2.29$	15.62	$+0.87$	$+0.0108$
2.0%	15.55	$+4.02$	17.56	$+2.81$	$+0.0197$
3.0%	18.86	$+7.32$	21.52	$+6.78$	$+0.0417$
5.0%	23.70	$+12.17$	26.61	$+11.86$	$+0.1013$

Note: Absolute poverty against baseline-fixed thresholds, before (BHC) and after (AHC) housing costs. Gini refers to equivalised household net income; baseline value 0.3641. Rates above 0.77% raise net revenue that is not recycled in the simulation. *Source:* Authors' calculations using PolicyEngine UK.

4.7 Validation and robustness

The swap has no benefit interactions. Running the reform through the full tax and benefit model and comparing with the arithmetic $\Delta y_i = CT_i - rL_i$, we find the two agree to within £0.25 for every household at every rate tested (0.5, 1.0 and 2.0 per cent), and the share of households for which they differ by more than £1 is zero. Abolishing council tax and levying a land tax changes no means-tested entitlement in the UK system: council tax reduction is itself abolished with the tax, and neither instrument enters the income or capital tests for Universal Credit, Pension Credit or the legacy benefits. The consequence is that the microdata, not the

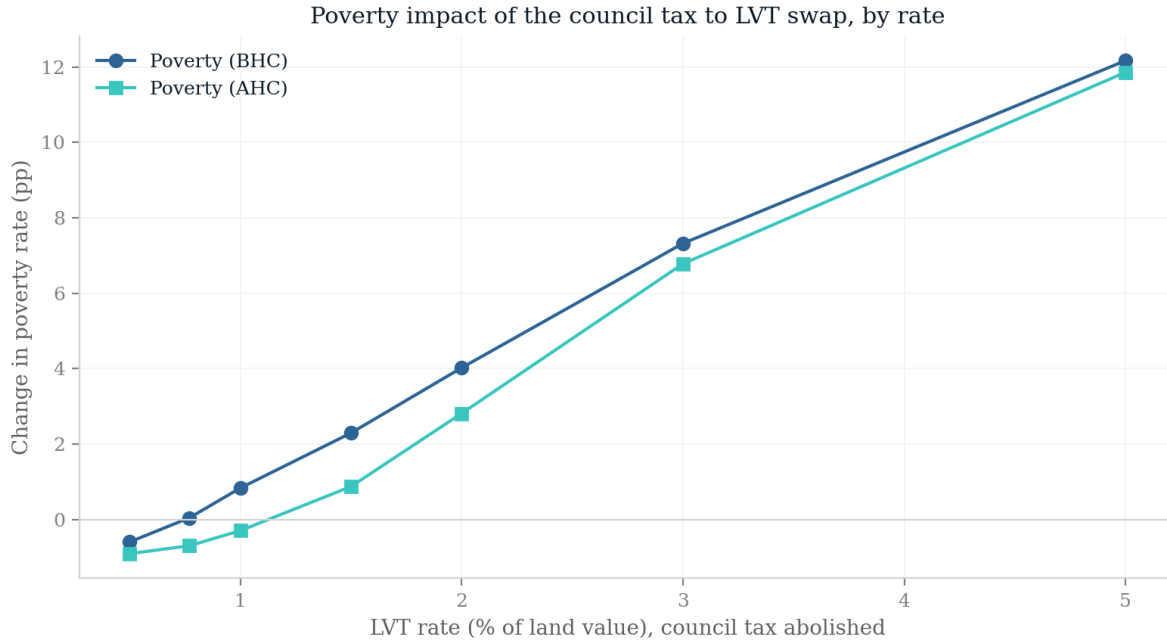


Figure 8: Poverty and Gini changes across the LVT rate grid.

tax-benefit calculator, do all the work in this reform — the calculator’s contribution is the council tax baseline, the equivalisation and the statutory poverty thresholds. We regard this as worth stating explicitly because it is easy to assume otherwise, and it means the results can be reproduced and audited without re-running the model.

Robustness. Table 7 varies the replacement target, the composition of the base and the instrument. The qualitative results are stable. The share of households gaining stays within a 65–70 per cent band across every specification. The budget-neutral rate ranges from 0.72 to 1.07 per cent, the top of that range being the household-land-only base, which is 28 per cent smaller. Calibrating the replacement target to OBR receipts rather than modelled liabilities lowers the rate to 0.72 per cent and, mechanically, makes every group better off — it is a £3.9 billion net tax cut rather than a neutral swap, and we therefore treat it as a bound rather than a competing central case. Excluding Northern Ireland, whose households would otherwise pay the land tax on top of domestic rates, raises the rate to 0.79 per cent and leaves the distribution essentially unchanged. Assuming half of corporate equity is foreign-owned and therefore outside the household base raises the required rate to 0.90 per cent, the largest single sensitivity in the table, but still leaves 68.0 per cent of households gaining. Perturbing every regional land share by ± 10 per cent, or rescaling household land to the un-uprated ONS 2024 benchmark, changes the rate by at most 0.06 percentage points and the decile effects by tens of pounds.

The final row is a comparator rather than a robustness check: a proportional tax on total property value, structures included, at the 0.73 per cent rate that raises the same revenue. It is less progressive on the wealth ranking — the top wealth decile loses £4,217 rather than £5,752 — and it produces fewer winners (64.4 against 68.2 per cent), because the base is spread more evenly across the property-owning population instead of being concentrated where location value

is highest. That difference is the quantitative content of the land/structures distinction.

Table 7: Robustness: revenue-neutral rate and distributional summary under alternative assumptions

Specification	Base (£tn)	Rate (%)	Winners (%)	Decile 1 (£)	Decile 10 (£)	Wealth 10 (£)	Δ Poverty AHC (pp)
Central	7.46	0.77	68.2	+647	−963	−5,752	−0.70
OBR receipts target	7.46	0.72	69.1	+698	−730	−5,177	−0.73
Great Britain only	7.34	0.79	67.9	+639	−1,016	−5,747	−0.56
Household land only	5.40	1.07	69.6	+574	−688	−5,364	−0.56
Half of corporate land foreign-owned	6.43	0.90	68.0	+617	−848	−5,589	−0.59
Household land scaled to ONS 2024	7.10	0.81	68.2	+651	−977	−5,772	−0.71
Land shares −10%	6.92	0.83	68.2	+653	−985	−5,782	−0.70
Land shares +10%	8.00	0.72	68.2	+642	−945	−5,726	−0.56
<i>Comparator:</i> proportional property tax	7.91	0.73	64.4	+605	−674	−4,217	−0.76

Note: Each row sets the rate so that revenue equals the replacement target on that row’s base. “Decile 1” and “Decile 10” are average net income changes by income decile; “Wealth 10” is the top wealth decile. *Source:* Authors’ calculations using PolicyEngine UK; `analysis/extensions.py`.

Capitalisation. The central results are flow accounting and assume no capitalisation. If instead the levy is fully anticipated and permanent, it capitalises into land prices: a rate r discounted at ρ implies a proportional fall of r/ρ in land values. At $\rho = 5$ per cent the 0.77 per cent rate implies a 15.4 per cent fall in land prices and an aggregate wealth loss of £1.15 trillion; at $\rho = 3$ per cent, a 25.7 per cent fall and £1.92 trillion. This is a transfer to the state of an asset value that was itself partly the capitalisation of past public investment, and it lands overwhelmingly on existing owners at announcement rather than on subsequent purchasers. Table 9 in the appendix reports the implied one-off wealth loss by wealth decile: at $\rho = 3$ per cent the top wealth decile loses £280,331 of land value, 6.5 per cent of its total wealth, against £323 for decile 3. Two points follow. First, the wealth-decile gradient of the reform is steeper still on a stock than on a flow basis. Second, the flow results in Sections 4.3–4.4 should be read as impact incidence in the first years, before the announcement effect on prices works through.

5 Discussion

5.1 Summary of findings

Our simulations indicate that a revenue-neutral replacement of council tax with a 0.77 per cent flat land value tax would leave 68 per cent of households better off, reduce absolute poverty after housing costs by 0.70 percentage points, and shift the tax burden substantially up the wealth distribution, while leaving before-housing-cost poverty and the Gini coefficient of equivalised household net income essentially unchanged. The dissociation between the wealth-decile and income-Gini results is the paper’s main analytical point, and it generalises: distributional statistics computed over the income distribution can be nearly uninformative about a reform whose incidence is organised by wealth. An LVT is a tax on a stock that is highly concentrated — we estimate that the top wealth decile holds 47.4 per cent of all land, and the wealth Gini is 0.70 against 0.36 for income — assessed on households whose position in the income distribution is

only weakly related to that stock. A reader who evaluated this reform only on the Gini and the BHC poverty rate, the two statistics that dominate UK distributional debate, would conclude that it does almost nothing. Table 5 shows that it does a great deal.

We are correspondingly cautious about the income-decile profile. Once the top two deciles are set aside, the average effects in deciles 2–8 are a few hundred pounds a year on incomes of £31,000–£69,000, they are not monotonic in income, and they rest on imputed land values whose decile means are visibly noisy. We do not think the data support claims about which particular middle decile gains; they support the claim that the reform is roughly neutral across the bottom eight deciles and costly to the top two.

The efficiency case, which our static framework does not quantify, points in the same direction as the equity case. Whereas council tax is levied on the value of dwellings, including improvements, an LVT imposes no charge on constructing, extending or upgrading property, and its base is immobile (George, 1879; Oates and Schwab, 1997).

5.2 Limitations

Several caveats qualify the results.

Static incidence. We assign the full burden to current landowners, with no behavioural response and no capitalisation in the central results. Standard theory predicts that an anticipated recurrent land tax capitalises into lower land prices (Feldstein, 1977), so part of the burden would fall on owners as a one-off wealth loss at announcement rather than as a permanent flow; conversely, abolishing council tax would be expected to capitalise into higher property prices. Section 4.7 quantifies the first of these: full capitalisation at discount rates of 3–5 per cent implies a fall in land values of 15–26 per cent, or £1.15–1.92 trillion, concentrated on the top wealth decile. Our flow-based accounting is best interpreted as the impact incidence in the initial years of the reform, and it understates how sharply the reform bears on land wealth.

Regional land shares. Equation (2) applies a single land share λ_r to every property in a region. In practice land shares vary within regions — they are higher in urban centres and lower for new-build dwellings on inexpensive land — so we are likely to understate within-region heterogeneity in LVT liabilities, although decile-level averages should be less affected.

Corporate land allocation. We allocate £2.06 trillion of corporate land to households in proportion to their measured corporate wealth. This abstracts from foreign ownership of UK corporate equity, which would shift part of the burden abroad and render the reform more favourable to UK households than we report, and it assumes full pass-through of the corporate-sector tax to shareholders rather than to workers or consumers.

Northern Ireland. Northern Ireland levies domestic rates rather than council tax; the central specification leaves rates unchanged, so Northern Irish households pay LVT in addition to their existing property tax. Section 4.7 reports the Great Britain-only alternative, which raises the required rate to 0.79 per cent and leaves the distributional pattern intact. A comprehensive UK-wide reform would fold domestic rates into the replacement, lowering the required rate slightly below our central figure.

Measurement. Land values are imputed rather than observed: Wealth and Assets Survey property wealth is self-reported, the upper tail of the survey remains thin even after enhancement, and the National Balance Sheet’s land estimates rest on the ONS’s residual valuation method (ONS, 2025). Council tax liabilities are modelled from regional band averages rather than parcel-level records.

5.3 Policy implications and design options

The most salient losing group comprises asset-rich, income-poor households — typically older owner-occupiers of valuable homes, concentrated in London and the South East — for whom an annual charge of 0.77 per cent of land value may represent a substantial share of cash income. Established remedies address this liquidity problem: deferral of the liability, with interest, until sale or death, as operated for property taxes in several US states and recommended by Mirrlees et al. (2011) and Muellbauer (2018); a phased transition over an extended period; or an exempt band of land value per household. Each option sacrifices some revenue or efficiency, but the underlying difficulty is one of liquidity rather than solvency, since the households concerned are by construction wealthy. The appendix quantifies the first option: households containing someone of state pension age owe 35.1 per cent of the total yield, £20.3 billion, so a universal pensioner deferral would postpone roughly a third of first-year receipts. That is the scale of the transitional financing requirement — large, bounded, and a cash-flow cost rather than a revenue loss, since deferred liabilities are recovered with interest from estates.

An alternative reform family would replace council tax with a proportional tax on total property value (structures plus land), as proposed by Muellbauer (2005) and modelled by Corlett and Gardiner (2018). A proportional property tax shares the LVT’s correction of council tax’s regressivity in value and obsolete valuations, and is administratively simpler because market prices are observed; however, it retains the disincentive to improve property and taxes the elastic (structures) margin alongside the inelastic one. We simulate this comparator directly in Section 4.7. At the 0.73 per cent rate that raises the same £57.6 billion, a proportional property tax produces fewer winners (64.4 against 68.2 per cent) and a shallower wealth gradient: the top wealth decile loses £4,217 rather than £5,752. The land base is more concentrated than the property base because location value is more concentrated than structure value, so for a given revenue target the LVT asks more of the largest landholders and less of everyone else. That is the quantitative content of the land/structures distinction, and it is the price paid for the proportional property tax’s administrative advantage of observable market prices.

5.4 Conclusion

This paper has presented, to our knowledge, the first open-source household-level microsimulation of a revenue-neutral replacement of council tax with a land value tax for the United Kingdom. At the budget-neutral rate of 0.77 per cent, our results suggest that the reform would benefit roughly two-thirds of households, reduce after-housing-cost poverty modestly, and reallocate the burden of recurrent property taxation from council tax’s regressive banded structure to the distribution of land wealth, with the largest losses concentrated in the top wealth decile. Measured income inequality and before-housing-cost poverty would be essentially unaffected — a result that underscores the importance of examining incidence along the wealth as well as the income distribution, and a caution against reading the near-zero movement in headline statistics as evidence that little has changed. The principal obstacles to implementation are practical

rather than analytical: valuing land separately from structures at scale, and managing the transition for asset-rich, income-poor owners. Both problems are long-standing, and both admit established policy responses. Extending the analysis to incorporate capitalisation, behavioural responses and parcel-level land valuation remains a subject for future research.

References

- Adam, S., Hodge, L., Phillips, D. and Xu, X. (2020). Revaluation and reform: bringing council tax in England into the 21st century. IFS Report R168, Institute for Fiscal Studies, London.
- ap Gwilym, R., Jones, E. and Rogers, H. (2020). A technical assessment of the potential for a local land value tax in Wales. Bangor University, report for the Welsh Government, Cardiff.
- Arnott, R. J. and Stiglitz, J. E. (1979). Aggregate land rents, expenditure on public goods, and optimal city size. *The Quarterly Journal of Economics*, 93(4), 599–616.
- Bonnet, O., Chapelle, G., Trannoy, A. and Wasmer, E. (2021). Land is back, it should be taxed, it can be taxed. *European Economic Review*, 134, 103696.
- Corlett, A. and Gardiner, L. (2018). Home affairs: options for reforming property taxation. Resolution Foundation, London.
- Dye, R. F. and England, R. W. (eds.) (2009). *Land Value Taxation: Theory, Evidence, and Practice*. Lincoln Institute of Land Policy, Cambridge, MA.
- Feldstein, M. (1977). The surprising incidence of a tax on pure rent: a new answer to an old question. *Journal of Political Economy*, 85(2), 349–360.
- George, H. (1879). *Progress and Poverty: An Inquiry into the Cause of Industrial Depressions and of Increase of Want with Increase of Wealth*. D. Appleton and Company, New York.
- Murphy, L. and Snelling, C. (2019). A poor tax: reforming council tax in London — final report. Institute for Public Policy Research, London, May 2019.
- Fairer Share and WPI Economics (2021). The proportional property tax: a fairer way to tax property in Britain. Fairer Share, London.
- Hughes, C., Sayce, S., Shepherd, E. and Wyatt, P. (2018). Investigation of potential land value tax policy options for Scotland: final report. Scottish Land Commission, Inverness.
- Johansson, Å., Heady, C., Arnold, J., Brys, B. and Vartia, L. (2008). Taxation and economic growth. OECD Economics Department Working Paper No. 620, OECD Publishing, Paris.
- Leunig, T. (2024). A fairer property tax. Onward, London.
- Ministry of Housing, Communities and Local Government (2023). Land value estimates for policy appraisal 2023. MHCLG, London.
- Mirrlees, J., Adam, S., Besley, T., Blundell, R., Bond, S., Chote, R., Gammie, M., Johnson, P., Myles, G. and Poterba, J. (2011). *Tax by Design: The Mirrlees Review*. Oxford University Press, Oxford.

- Muellbauer, J. (2005). Property taxation and the economy after the Barker Review. *The Economic Journal*, 115(502), C99–C117.
- Muellbauer, J. (2018). Housing, debt and the economy: a tale of two countries. *National Institute Economic Review*, 245(1), R20–R33.
- Oates, W. E. and Schwab, R. M. (1997). The impact of urban land taxation: the Pittsburgh experience. *National Tax Journal*, 50(1), 1–21.
- Office for Budget Responsibility (2025). Economic and Fiscal Outlook, November 2025. OBR, London.
- Office for National Statistics (2025). National balance sheet estimates for the UK: 2025. ONS, Newport. <https://www.ons.gov.uk/economy/nationalaccounts/uksectoraccounts/bulletins/nationalbalancesheet/2025>.
- PolicyEngine (2026). PolicyEngine UK: an open-source microsimulation model of the UK tax and benefit system. <https://policyengine.org/uk>.
- OECD (2024). OECD Economic Surveys: United Kingdom 2024. OECD Publishing, Paris.
- Ramsey, F. P. (1927). A contribution to the theory of taxation. *The Economic Journal*, 37(145), 47–61.
- Tideman, N. (1995). Taxing land is better than neutral: land taxes, land speculation and the timing of development. Lincoln Institute of Land Policy Working Paper, Cambridge, MA. Reprinted in K. C. Wenzer (ed.), *Land-Value Taxation*, M. E. Sharpe, 1999.
- Commission on Local Tax Reform (2015). Just change: a new approach to local taxation. Scottish Government, Edinburgh.

A Appendix

A.1 Impact by income decile at alternative rates

Table 8 reports the average net income change by income decile across the full rate grid, with council tax abolished in every scenario. At 0.5 per cent every decile gains on average, the reform reducing net revenue by £20.3 billion; from 1.5 per cent upwards every decile loses on average, with losses concentrated in deciles 9–10. Rates other than 0.77 per cent are not revenue-neutral, so these columns combine the incidence of the swap with that of a net tax cut or rise.

A.2 Landless households

Households holding no land — neither residential property nor corporate wealth — make up 17.2 per cent of the population. Because their LVT liability is zero at any rate, their outcome is invariant across the rate grid: an average gain of £832 per year (their abolished council tax bill), with 81.5 per cent winning. The non-winners are predominantly households whose council tax was already fully covered by council tax reduction, for whom abolition changes nothing.

Table 8: Average net income change (£/year) by income decile and LVT rate

Decile	0.5%	0.77%	1.0%	1.5%	2.0%	3.0%	5.0%
1	+907	+647	+430	−46	−523	−1,476	−3,382
2	+430	−163	−659	−1,747	−2,835	−5,012	−9,364
3	+893	+578	+315	−264	−842	−1,999	−4,313
4	+491	−60	−520	−1,532	−2,543	−4,566	−8,611
5	+682	+145	−304	−1,290	−2,276	−4,247	−8,190
6	+753	+179	−300	−1,353	−2,406	−4,511	−8,722
7	+963	+455	+31	−901	−1,833	−3,697	−7,426
8	+798	+35	−603	−2,004	−3,405	−6,208	−11,812
9	+132	−1,016	−1,976	−4,084	−6,192	−10,409	−18,842
10	+243	−963	−1,972	−4,186	−6,401	−10,830	−19,689

Note: Rates above 0.77% raise net revenue that is not recycled; rates below it reduce net revenue. *Source:* Authors' calculations using PolicyEngine UK.

A.3 Worked single-household example

Consider a household owning a £400,000 home in London, where the land share is 85 per cent, implying household land of £340,000. If the household also holds £50,000 of corporate wealth, its attributed corporate land is that sum multiplied by the population ratio of aggregate corporate land to aggregate corporate wealth, which the simulation puts at £0.145 per pound of corporate wealth, or £7,257. Its total land value is therefore £347,257 and its annual liability at the budget-neutral rate is $0.0077 \times 347,257 = £2,682$; at a 1 per cent rate it would be £3,473. Whether the household gains depends on the abolished council tax bill: with a typical upper-band London bill of around £2,400 it loses roughly £280 a year, and it breaks even at a bill of £2,682.

The allocation ratio must be computed against the weighted population. Evaluating equation (2) inside a single-household simulation assigns the entire £2.06 trillion corporate land base to that one record, because it constitutes the whole population; the example above therefore uses the population ratio from the microsimulation.

A.4 Capitalisation by wealth decile

Table 9 reports the one-off wealth loss implied by full capitalisation of the permanent 0.77 per cent levy, $\Delta W_i = -(r^*/\rho)L_i$, at three discount rates. The aggregate loss is £1.15 trillion at $\rho = 5$ per cent, £1.44 trillion at 4 per cent and £1.92 trillion at 3 per cent, corresponding to falls in land prices of 15.4, 19.3 and 25.7 per cent. Expressed as a share of total wealth the loss peaks in the middle of the wealth distribution — deciles 5 and 6, whose wealth is overwhelmingly housing — while in absolute terms it is concentrated at the top.

A.5 Deferral for pensioner households

The liquidity problem the reform creates is concentrated and measurable. Households containing at least one person of state pension age make up 27.2 per cent of the population and would owe £20.3 billion of the £57.6 billion total, or 35.1 per cent of the yield, at an average liability of £2,318. A full deferral option for pensioner households — the liability rolled up with interest against the property, as recommended by [Mirrlees et al. \(2011\)](#) — would therefore postpone

Table 9: Capitalised one-off wealth loss by wealth decile (£), 0.77% rate

Wealth decile	$\rho = 3\%$		$\rho = 5\%$	
	Loss (£)	% of wealth	Loss (£)	% of wealth
1	0	—	0	—
2	8	2.9	5	1.7
3	323	3.6	194	2.2
4	6,502	8.0	3,901	4.8
5	19,746	9.7	11,848	5.8
6	35,785	9.4	21,471	5.6
7	54,668	8.3	32,801	5.0
8	82,760	7.9	49,656	4.7
9	116,209	6.8	69,725	4.1
10	280,331	6.5	168,199	3.9

Note: Loss is $(r^*/\rho)L_i$ averaged within decile; “% of wealth” is that loss over average total wealth. Deciles 1–2 hold negligible land. *Source:* Authors’ calculations; `analysis/extensions.py`.

rather than forgo roughly a third of receipts in the first year, declining thereafter as deferred estates are settled. This is the scale of the transitional financing problem; it is large but bounded, and it is a cash-flow cost to government rather than a revenue loss.

A.6 Data and parameter summary

A.7 Replication note

All results are generated from a single baseline simulation of PolicyEngine UK 2.88.20 on the pinned Enhanced FRS release `enhanced_frs_2023_24@1.55.10`, together with the closed-form arithmetic validated in Section 4.7. The generating script (`uk_lvt/pipeline_direct.py`) re-runs the full model at three rates on every invocation and refuses to write results if the arithmetic and the model disagree by more than £1 per household or 0.01 percentage points of poverty.

Two caveats on version sensitivity are worth recording. First, poverty *levels* in this model move with the underlying benefit and uprating parameters: an earlier release (2.88.10) put baseline absolute BHC poverty at 12.47 rather than 11.53 per cent, and the estimated change at the budget-neutral rate at -0.65 rather than $+0.04$ percentage points. The revenue, land and winner-share results are unaffected, but the sign of the BHC poverty effect is not robust across model versions, which is why we do not report a poverty reduction on that measure. Second, the wealth-decile results depend on how deciles are constructed in the presence of tied zero-wealth households; we construct equal-weight deciles directly rather than using the model’s own decile variable.

Table 10: Key data sources and parameters

Item	Value	Source / method
Microdata	EFRS 2023–24	FRS + SPI quantile-forest enhancement
Model version	policyengine-uk 2.88.20	Pinned; see replication note below
Simulation year	2026	PolicyEngine UK
Households	32.08m	Weighted EFRS household count
Council tax, gross	£61.9bn	Modelled liabilities, VOA band calibration
Council tax reduction	£4.3bn	Modelled
Council tax, net (R_{CT})	£57.6bn	Equation (1)
Memo: OBR receipts	£53.7bn	OBR EFO; robustness row
Household land	£5.40tn	Model output, equation (2)
Corporate land	£2.06tn	ONS NBS aggregate; allocated by corporate wealth
Total land base (L)	£7.46tn	105.1% of un-uprated ONS 2024 (£7.10tn)
Regional land shares	42%–85%	MHCLG residual method, equation (3)
Budget-neutral rate (r^*)	0.77%	R_{CT}/L
Reform parameters	<code>lvt.rate</code> , <code>abolish_ct</code>	<code>gov.contrib.ubi_center.land_value_tax.rate</code> ; <code>gov.contrib.abolish_council_tax</code>